1st task

A) Dogbert measured the river depth at several locations. Find the largest depth value that is smaller than the depth measured immediately after it. If there is no one, print out –1.

1st example input:
number of locations: 8
depth: [2,4,2,5,6,2,1,7]

Output: 5

2nd example input
number of locations: 4
depth: [2,2,2,2]

Output: -1

(solution: https://progalap.elte.hu/specifikacio/v1/?uuid=d7a950c3-60e0-4ae8-99ba-c51c43917331)

B) Dilbert recorded the price of dog meals over several days. Was there continuous inflation during this period? ('Continuous inflation' means that the price increased every single day.)

1st example input:
number of days: 6
prices: [1,2,3,7,8,9]

Output: true

(solution: https://progalap.elte.hu/specifikacio/v1/?uuid=c31d8332-47b9-428c-b9fa-1b85d8e3c9f1)

C) Dilbert recorded Dogbert's runs, including the time and the location he ran to. List the locations for which Dogbert ran for more than 30 minutes.

1st example input:
number of runs: 4
runs:

time	*location*
15	office
35	restaurant
40	gym
5	shop

Output: [restaurant, gym]

(solution: https://progalap.elte.hu/specifikacio/v1/?uuid=6eebda91-0a9a-4d73-8295-71f6359ca42d)

D) Dilbert recorded Dogbert's walks, including the distance and the time. Did Dogbert walk faster than 4 km/h? (v=s/t)

1st example input:
number of walks: 4
runs:

distance	*time*
15	2.5
35	3.5
40	5.2
5	2.1

Output: true

(Solution: https://progalap.elte.hu/specifikacio/v1/?uuid=d337338b-27c7-4cd0-bd42-da24a3c1da06)

2nd task

A) Catbert recorded the working hours of 'm' colleagues over 'n' days. Find any colleague whose average working time is greater than 'k' hours.

Example input:
n: 5, m: 3, k: 8

Output: 2

	1.coll.	2.coll.	3.coll.
1.day	12	2	0
2.day	4	5	1
3.day	1	10	2
4.day	0	12	3
5.day	1	14	1

(Solution: https://progalap.elte.hu/specifikacio/v1/?uuid=eaa64c41-540a-41aa-a687-6a5626203724)

B) Catbert eats five meals a day. She prepared a table to record the calories for each meal. List all the meals (index of meals) with the smallest calorie value that is still greater than 'k' (in kcal).

Example input:
Number of days: 4, k: 3

Output: [2, 3, 5]

	1. meal	2. meal	3. meal	4. meal	5. meal
1. day	12	12	10	5	5
2. day	4	5	11	12	23
3. day	1	10	12	2	4
4. day	2	5	21	4	7

(solution: https://progalap.elte.hu/specifikacio/v1/?uuid=0982061b-020c-429f-a768-30892d92efb1)

C) Catbert recorded the break length of 'n' colleagues over 'm' days. How many colleagues could she find who had a break that was longer than their break on the previous day?

Example input:
n: 4, m: 5

Output: 2

	1.day	2.day	3.day	4.day	5.day
1.coll.	12	12	10	5	2
2. coll.	4	5	1	12	23
3. coll.	1	10	2	4	2
4. coll.	2	2	2	2	2

(Solution: https://progalap.elte.hu/specifikacio/v1/?uuid=1814df8a-bb55-41a0-b47f-b3de0ac38b2f)

D) Catbert organised a multi-day conference with four coffee breaks each day. She recorded how much coffee was consumed during each break. Find a day on which the participants drank more than 5 litres of coffee in each of the coffee breaks.

Example input:
Number of days: 4

Output: 3

	1. break	2. break	3. break	4. break
1. day	12	12	10	5
2. day	4	5	11	12
3. day	11	10	12	12
4. day	12	5	2	4

(solution: https://progalap.elte.hu/specifikacio/v1/?uuid=2618df0d-65e1-470d-8799-232eb5084cca)

3rd task

Ratbert counted how many times each keyword appears in a code. List the keywords whose counts are greater than the largest count of any keyword that comes before them in the list.

1st example input:
number of keywords: 5
keywords:

word	How many times…
program	1
if	10
else	5
while	20
do	12

Output: [if, while]

Solution: https://progalap.elte.hu/specifikacio/v1/?uuid=61208a48-b7f1-4b70-97a0-eabdcf81a9f1 , 2nd sol: dec.all instead of max)

B) Ratbert recorded the size of each rat family and the name of the location where they live. Give a family whose size is greater than the largest family size among all families that appear before it in the list.

1st example input:
number of families: 5
data:

size	location
10	Brüssel
100	Budapest
55	Milan
210	London
120	Prague

Output: true, 2

Solution: https://progalap.elte.hu/specifikacio/v1/?uuid=a4a15202-20ac-4c02-aa38-18e05d403523

C) Ratbert generated several random numbers between 'a' and 'b'. For each distinct number in the interval a..b list how many times it appears in the input!

Example input:
a: 1, b: 6
numbers: [1,2,4,2,5,1,2,4,1,5]

Output: [3,3,0,2,2,0]

(Solution: https://progalap.elte.hu/specifikacio/v1/?uuid=df8bfb14-d641-48dc-b6e8-ff9c5fe98cfc)

D) Ratbert recorded the house numbers he visited on the Rectangle Square over 'n' days. List the houses that were visited more than 'k' times. On the square there are 6 houses at all.

Example input:
k: 2, number of days: 10,
numbers: [3,4,2,6,3,2,4,6,4,2]

Output: [4,2]
or [2,4] - it depends on your solution

Solution:
https://progalap.elte.hu/specifikacio/v1/?uuid=2fc44c96-5255-4875-bc5b-6ef681fc896e

https://progalap.elte.hu/specifikacio/v1/?uuid=aacdde10-3aae-4a21-8506-70c83fa44bd8